

RESEARCH INTERESTS

- **Area:** Statistical genetics; Post-GWAS(Genome-Wide Association Studies); Pleiotropy of complex diseases.
- **Methodology:** Statistical learning, deep learning, and AI method/tool development for GWAS/Omics data.
- **Applications:** Adolescent mental health, war-related trauma, and family mental health interactions.

EDUCATION

Expected 05/2027	Brown University Sc.M. in Biostatistics GPA: 4.0/4.0	Thesis Advisor: Dr. Ying Ma
09/2021–07/2025	University of Nottingham, Ningbo Campus (UNNC) B.S. (Honors) in Mathematics with Applied Mathematics Undergraduate Dissertation: Bifurcation and Complex Dynamics in a Discrete Predator–Prey Model with a Prey Allee Effect Dissertation Advisors: Dr. Elio Eduardo Espejo Arenas and Dr. Mainul Haque	

PUBLICATIONS

Fan Z., Gao B.*, & Ma Y.* (2026). **Graph-Regularized Sparse Factorization Identifies Phenome-Wide Pleiotropic Genetic Factors**

Manuscript in preparation; bioRxiv preprint planned for December 2026.

*Corresponding authors.

- **Research Question:** Why do different complex traits share genetic associations, and can we recover the underlying shared genetic structure from large collections of GWAS summary statistics?
- **Key Findings:** We proposed a functional annotation-regulated factorization method FLARE, FLARE recovers genetic factors that replicate across biobanks and ancestries and map onto recognizable biology.
- **Innovation:** Integrate functional genomic annotations into graph-regularized sparse factorization to model shared genetic architecture across traits while retaining interpretable, biologically informed latent factors.
- **My Contributions:**
 - **Developed the FLARE algorithm in R/Rcpp:** Constructed the graph annotation matrix, implemented model fitting using proximal gradient methods/FISTA and glmnet, and performed model selection; maintained the code in Git to support reproducibility and collaboration.
 - **Designed and conducted extensive model-based and real-data-based simulations:** Evaluated false discovery using null-GWAS and over-specified-rank designs, and constructed semi-real benchmarks from All of Us to compare FLARE with GFA, GLEANR, FactorGo, flash, and SVD.
 - **Designed and conducted diverse real-data analyses:** Performed factorizations across UK Biobank, FinnGen, All of Us, and Biobank Japan; evaluated cross-biobank replication across independent European and East Asian cohorts; and conducted downstream analyses including factor-level GWAS, genome-wide stratified LD score regression using tissue chromatin annotations, and enrichment analyses validated against loading-permutation nulls.

Fan, Z., Rajasankar, V., Dunsiger, S., Panok, V., Burlaka, V., Serdiuk, O.*, & Sun, S.* (2026). **War Exposure and Psychological Distress in Ukrainian Adolescent-Caregiver Dyads**

Manuscript in preparation; medRxiv preprint planned for November 2026.

*Corresponding authors.

- **Research Question:** What is the association between adolescent war exposure and psychological distress in Ukrainian adolescent-caregiver dyads?
- **Key Findings:** In this cross-sectional study of 1,604 Ukrainian adolescent-caregiver dyads, adolescent war exposure was associated with elevated depressive and anxiety symptoms in adolescents but was not significantly associated with caregiver symptoms. Adjusted adolescent-caregiver symptom associations were positive, and statistically significant.
- **Meaning:** These findings highlight the importance of considering both adolescent and caregiver mental health and developing family-centered support for war-affected Ukrainian youth.
- **My Contributions:**
 - **As lead analyst, designed and implemented the statistical analysis:** Developed a detailed cross-sectional analysis plan and specified study measures and outcomes. Used logistic generalized estimating equations (GEE) for dyadic analyses of adolescent-caregiver mental health associations in the context of war exposure.

- **As a main contributor, designed and implemented data cleaning and linkage:** Linked datasets comprising 2,200+ adolescents and 1,800+ caregivers to prepare the dyadic analytic dataset. Developed a detailed data-cleaning protocol and prepared separate adolescent, caregiver, dyadic, and sensitivity-analysis datasets for team use.
- **Coordinated with the Ukrainian collaborating principal investigator:** Represented the research team in discussions to clarify and confirm dataset and study design details.

RESEARCH EXPERIENCE

08/2026–Present **Statistical method development for localizing trait-associated tissue regions by integrating GWAS with spatial transcriptomics**

- **Developing the GWAS analysis pipeline:** preparing GWAS summary statistics and LD reference panels, and deriving gene-level association statistics with MAGMA as genetic input to the method.
- **Contributing to statistical method development:** working with a senior PhD candidate on a framework that integrates GWAS-derived genetic evidence with spatial gene expression to localize trait-associated tissue regions.
- **Benchmarking on real spatial transcriptomics datasets:** running scDRS and gsMap as competing methods and comparing their spatial localization of trait-relevant signals with ours.

*Collaborating with Chichun Tan, The Ma Lab | Brown University;
Advised by Dr. Ying Ma, The Ma Lab | Brown University.*

10/2025–Present **Statistical method development for pleiotropy discovery from multi-biobank GWAS summary statistics**

- **Carrying the project end to end:** algorithm design and implementation, all simulations, all real-data analyses and the full manuscript, from an initial idea proposed by my advisors and under weekly advising.
- **Building reproducible HPC pipelines and project organization:** isolated, resumable Slurm workflows in R/Python/LDSC with frozen inputs, checksummed references and smoke tests before every large run.
- **Designing the method and benchmarking it against competing methods:** designed and implemented every comparison from simulation to real data, against seven existing methods (GLEANR, GFA, FactorGo, flash, SVD and two SVD variants) and a no-graph ablation.
- **Writing the manuscript and preparing figures:** drafted the full manuscript and produced all publication-quality visualizations.

*Advised by Dr. Ying Ma, The Ma Lab | Brown University;
Co-advised by Dr. Boran Gao | Purdue University.*

10/2025–Present **Dyadic and longitudinal modeling of war exposure and mental health in the Ukraine Youth and Family Longitudinal Study**

- **First-authoring one of the wave-1 manuscripts:** cross-sectional analysis of about 2,200 Ukrainian adolescents and 1,900 caregivers for wave 1 data on how war exposure and family-level factors relate to mental health outcomes.
- **Modeling adolescent-caregiver dyads with APIM-GEE:** actor and partner effects linking caregiver war exposure and mental health to adolescent depression, anxiety and related outcomes.
- **Extending to longitudinal analysis with wave 2 (2026):** building repeated-measures models to assess change in mental health over time.

Advised by Dr. Shufang Sun, Global Mental Health and Flourishing Lab | Brown University.

SKILLS

- **Computing & Programming:** R, Python, Linux, Git, PyTorch, C++, Bash/Zsh, Matlab, L^AT_EX, Markdown.
- **Statistical Methodology:** Inference Theory, Bayesian, Causal Inference, Linear/GLM, Machine Learning, Deep Learning.
- **Mathematics Theory:** Linear Algebra, Calculus/Vector Calculus, Differential Equations, Real Analysis, Numerical Analysis, Optimization, Graph Theory.
- **Statistical Genetics Methods:** GWAS summary-statistics analysis, LD score regression (LDSC, stratified/cell-type-specific LDSC, LD score computation), gene-based and gene-set analysis (MAGMA), matrix-factorization methods for GWAS (GLEANR, GFA, FactorGo, flash), integration of GWAS with single-cell and spatial transcriptomics (scDRS, gsMap), tissue enrichment analysis, variant-to-gene mapping.
- **Genomic Data & Resources:** UK Biobank, All of Us, FinnGen, Biobank Japan, 1000 Genomes reference panels (PLINK), GTEx, Roadmap/ENTEX chromatin annotations, Enrichr gene-set libraries, Open Targets; HPC with Slurm.

SELECTED HONORS & AWARDS

2025–2026 **Todd '87 and Christine Fisher P'17 Graduate Fellowship, Brown University**

2024 **Global Engagement Office Summer Exchange Tuition Waiver Scholarship, UNNC**

CONFERENCES

06/2026 **Oral presentation: Displacement and Mental Health Symposium; Yale University, New Haven, CT.**
Rajasankar V, Fan Z, Skeen SJ, Werner L, Go M, Giovannone N, Shebanov O, Burlaka V, Panok V, Dunsinger S, Serdiuk A, Sun S. Mental health, suicide risk and behavioral profiles among Ukrainian adolescents: Findings from Wave 1 of the Ukraine Longitudinal Study.

TEACHING EXPERIENCE

06/2024–07/2024 **Teaching Assistant, MATH3048: Mathematical Medicine and Biology, UNNC**

PROFESSIONAL EXPERIENCE

IQVIA plc **Shanghai, China**
11/2024–01/2025 **Part-time Analyst Intern, Management Consulting Department**

HALMA plc **Shanghai, China**
07/2024–10/2024 **Business Intelligence and M&A Research Summer Intern, Healthcare Sector**

Beijing AIMEDICINE Co., Ltd. × NYU Shanghai **Shanghai, China**
07/2023–08/2023 **Data Analyst, Yangtze River Delta Digital Elderly Care Innovation White paper Project**

LEADERSHIP, VOLUNTEERING & PUBLIC SERVICE

09/2023 **The 19th Asian Games, Volunteer**
04/2022–04/2023 **UNNC Campus Ambassador Association, Vice President**
06/2022 **UNNC Summer Social Practice, Volunteer Teaching Team Leader**

REFERENCES

Ying Ma Thesis & Research Advisor Assistant Professor of Biostatistics Brown University ying_ma@brown.edu	Boran Gao Research Advisor Assistant Professor of Statistics Purdue University gao824@purdue.edu
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Shufang Sun
Research Advisor (applied collaboration)
Associate Professor of Behavioral and Social Sciences
Brown University
shufang_sun@brown.edu